RP-Style Decision Circuit

Corrected mathematics & technical interpretation

A revised guide to the intended logic, independent-trial error bounds, photon-count statistics, and the limits of feedback analogies.

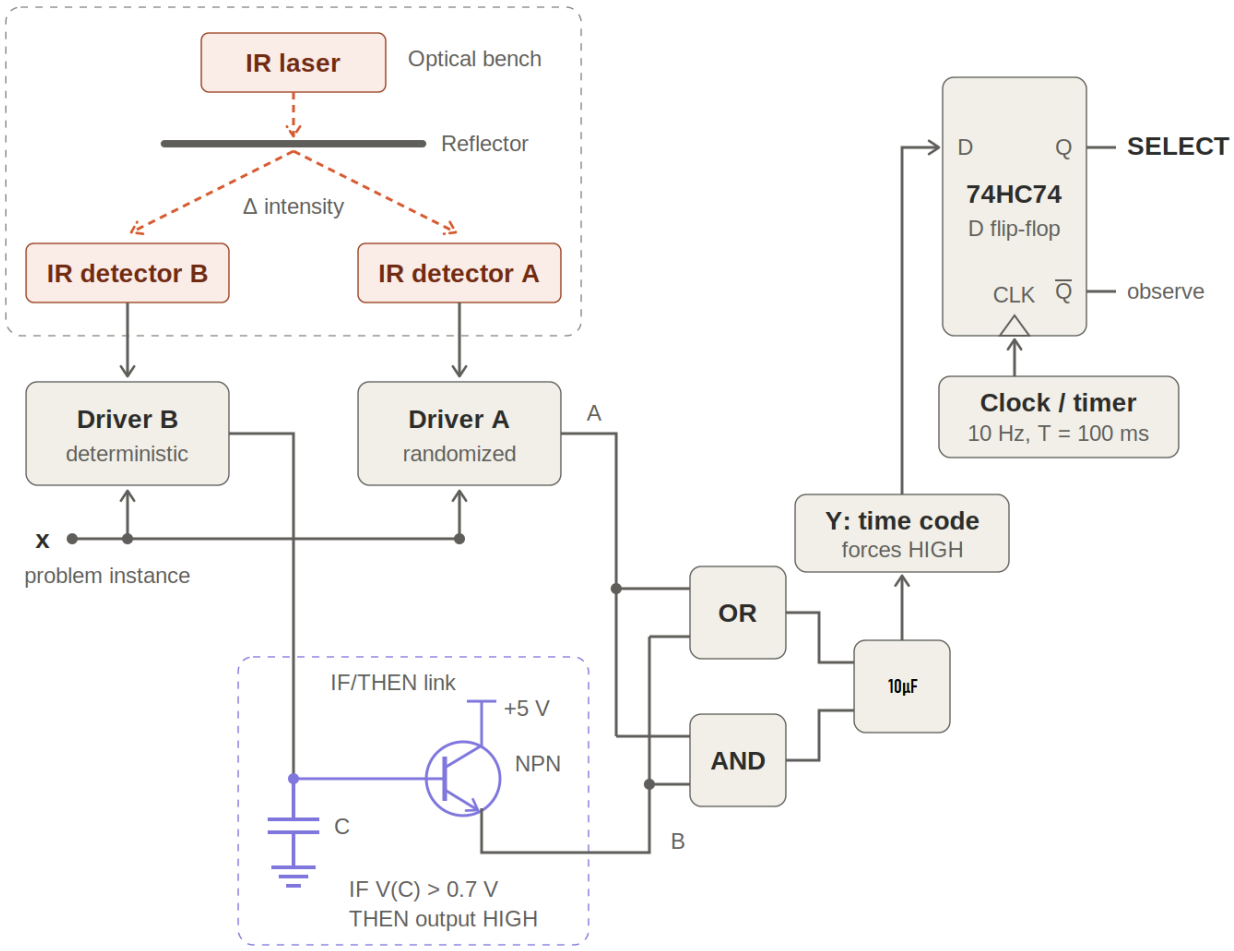


FIGURE 1 · Original conceptual circuit schematic, restored from the supplied PDF. Its wiring and labels are reproduced for reference, not validated as a construction-ready circuit. The corrected specifications on the following pages take precedence.

REVISED EDITION • Mathematics, assumptions, and interpretation corrected

Decision model and guarantees

The diagram proposes parallel randomized and deterministic drivers followed by logic, conditioning, and clocked memory. Its hardware labels do not by themselves establish an RP algorithm, nonclassical light, or a closed timelike curve.

The RP definition

Let L be a language, x an input, and r the random bits used by an algorithm M. One-sided error requires:

$$x \notin L \Rightarrow \Pr_r[M(x,r)=1]=0$$

$$x \in L \Rightarrow \Pr_r[M(x,r)=1] \geq \frac{1}{2}$$

The worst-case running time must also be polynomial in input length for every random choice. A fixed sampling frequency does not establish this scaling requirement.

Driver requirements

$$A = w(x,r), \qquad B = c(x), \qquad A, B \in \{0,1\}$$

$$D = (A \vee B) \wedge (A \wedge B) = A \wedge B$$

$$\Pr_r[D=1 \mid x] = c(x)\Pr_r[w(x,r)=1]$$

Driver A must meet the RP witness-test guarantees. Driver B must equal 1 on every YES instance; otherwise it can permanently veto valid inputs. The Boolean reduction assumes an explicit combining AND operation, not tied push-pull gate outputs.

Case	A	B	D
NO instance	0	0 or 1	0
YES, witness found	1	1 required	1
YES, witness missed	0	1 required	0

Timing, memory, and repeated trials

Enable valid trials; do not force acceptance

$$D_j = A_j \wedge B_j \wedge Y_j, \qquad (Y_1, Y_2, Y_3, Y_4) = (1, 0, 1, 1)$$

Time Code Y may mask a trial. It must not independently force D HIGH. OR-ing Y into the acceptance condition would allow NO instances to be accepted. Count only enabled, completed trials.

$$Q(t_j^{+}) = D(t_j^{-}), \qquad \overline{Q}(t_j^{+}) = 1 - Q(t_j^{+})$$

This ideal rising-edge rule assumes setup/hold requirements are met and asynchronous controls are inactive. The complementary output is not an independent observation channel.

Accept if any independent trial accepts

$$S_0 = 0, \qquad S_j = S_{j-1} \vee D_j$$

$$\Pr[S_k = 0 \mid x \in L] = \prod_{j=1}^{k} (1 - p_j) \le 2^{-k}$$

Here each independent trial has YES-instance success probability p_j ≥ 1/2. With a common success probability p, the rejection probability is (1 − p) to the power k. For NO instances, acceptance probability remains zero under the abstract guarantees.

The bound is “at most,” not necessarily equality. Simply overwriting a flip-flop with the most recent trial does not accumulate acceptance.

$$f = 10\,\mathrm{Hz}, \qquad T = 1/f = 0.1\,\mathrm{s}, \qquad t_k = kT$$

Trials	Rejection upper bound	Nominal time
1	0.5	0.1 s
2	0.25	0.2 s
5	0.03125	0.5 s
10	0.0009765625	1.0 s
20	0.00000095367431640625	2.0 s

Times assume one completed, enabled trial per period and no overhead. Disabled windows add elapsed time, not trials.

Photon-count statistics

Use Q for the flip-flop output and Q_M for the Mandel parameter. Photon-count moments must refer to the same specified mode or counting bin, with positive mean count.

$$\mu = \langle n \rangle > 0, \qquad F = \frac{\mathrm{Var}(n)}{\mu}, \qquad Q_{\mathrm{M}} = F - 1$$

$$g^{(2)}(0) = \frac{\langle n(n-1) \rangle}{\mu^2}$$

$$F = 1 + \mu[g^{(2)}(0) - 1]$$

$$Q_{\mathrm{M}} = \mu[g^{(2)}(0) - 1]$$

The relationship follows from expanding the factorial moment: the mean of n(n − 1) equals the variance plus the squared mean minus the mean. A finite-window count statistic is not automatically interchangeable with a differently defined time-resolved correlation measurement.

Ideal count model	Fano factor	g²(0)
Poisson distribution	1	1
Single-mode thermal distribution	1 + mean count	2
Exactly one photon in the mode	0	0

Interpretation boundaries

A Fano factor below one identifies sub-Poissonian counts in the specified counting model. A unit Fano factor alone does not prove the complete distribution is Poissonian. Thermal and Poisson models must not be treated as equivalent merely because the excitation is described as random.

Relabeling an NPN threshold circuit as regulated injection does not establish single-electron transport, zero carrier-count variance, or optical antibunching. These require a specified source model and measurements.

Carrier thinning and detection

State the independent-conversion assumption

$$N_{\text{ph}} \mid N_e \sim \text{Binomial}(N_e, \eta)$$

Each carrier independently yields at most one detected photon with the same probability eta, without added counts or correlations. For positive mean detected count:

$$\mathbb{E}[N_{\text{ph}}] = \eta \mathbb{E}[N_e]$$

$$\text{Var}(N_{\text{ph}}) = \eta(1-\eta)\mathbb{E}[N_e] + \eta^2 \text{Var}(N_e)$$

$$F_{\text{ph}} = 1 - \eta + \eta F_e = 1 - \eta(1 - F_e)$$

For F_e = 0 and eta = 0.8, the resulting F_ph is 0.2. This is a consequence of the stated model, not proof that the depicted hardware achieves F_e = 0.

Independent detection opportunities

$$P_{\text{miss},\,k} = (1-\eta)^k$$

This expression requires a new independent detection opportunity per trial. It does not describe repeated detection attempts on the same absorbed photon.

Dark counts break the literal RP analogy

$$P_{\text{false accept},\,k} = 1 - (1-d)^k \approx kd \quad (kd \ll 1)$$

Here d is the independent dark-click probability per window, with acceptance on any click. The expected dark-click total is kd, but the probability of at least one dark click is only approximately linear when kd is small. Nonzero dark clicks eliminate an exact zero-false-accept detector guarantee.

Transitions and measurement

Excitation and emission have opposite directions

$$E_1 \longrightarrow E_2 : \quad \text{excitation requires energy } E_2 - E_1$$

$$E_2 \longrightarrow E_1 : \quad h\nu = E_2 - E_1$$

Upward excitation requires supplied energy. Downward radiative decay emits a photon. The original switching table incorrectly associated emission with the upward transition.

Ideal coherent excitation

$$P_e = \sin^2(\theta/2), \qquad \theta = \pi \;\Rightarrow\; P_e = 1$$

This assumes an ideal resonantly driven two-level system initially in the ground state, neglecting decay and dephasing during the pulse. A classical driver labeled deterministic does not establish these physical conditions.

Number measurements are not binary clicks

$$P(n) = |\langle n|\psi\rangle|^2$$

This is the ideal number-measurement probability for a pure single-mode state. A binary click/no-click detector is not generally photon-number resolving; a click histogram is not automatically the complete photon-number distribution. Efficiency, dark counts, and detector response belong in the measurement model.

Do not equate unrelated time constants

Emitter lifetime, detector recovery time, and an external RC decay time are distinct physical quantities. A mapping between them needs a dynamical model; a common symbol does not establish equality.

Feedback and CTC models

Ordinary feedback is causal

Returning Q to an excitation driver can change future inputs. A feedback wire does not make that signal condition its past input or implement a closed timelike curve.

Deutsch consistency

$$\rho_{\mathrm{CTC}} = \mathrm{Tr}_{\mathrm{other}}\left[U(\rho_{\mathrm{other}} \otimes \rho_{\mathrm{CTC}})U^{\dagger}\right]$$

This equation specifies a consistency condition in a hypothetical CTC model. It is not derived from ordinary electronic feedback.

Classical and quantum NOT are different

$$p = 1 - p \quad \Rightarrow \quad p = \tfrac{1}{2}$$

$$\rho = X\rho X, \qquad \rho = \tfrac{1}{2}(I + r_x X + r_y Y + r_z Z)$$

$$r_y = r_z = 0, \qquad -1 \le r_x \le 1$$

Classical NOT has a consistent equal-probability distribution. The quantum NOT channel has a family of fixed points: I/2 is its maximum-entropy member, not its unique solution. These equal probabilities do not derive the RP acceptance threshold.

Computational power requires the resource assumption

$$\mathrm{P}_{\mathrm{CTC}} = \mathrm{BQP}_{\mathrm{CTC}} = \mathrm{PSPACE}$$

The Aaronson–Watrous theorem assumes access to Deutsch-style causal-consistency resources. Ordinary feedback circuits do not inherit this power. The theorem also does not establish a strict separation between RP and PSPACE.

Implementation notes and references

Electrical specifications still required

$$\tau = RC, \qquad f_c = \frac{1}{2\pi RC}$$

These expressions apply to a defined ideal first-order RC network. A 10 µF label alone does not specify settling time or cutoff frequency. Charging, loading, threshold crossings, discharge paths, and sampling margins require the actual circuit topology.

A base-emitter voltage near 0.7 V is an approximation, not a universal threshold. An emitter follower does not automatically drive its output to the collector supply. Do not directly tie push-pull OR and AND outputs together. Use selected component datasheets to specify voltage limits, inactive asynchronous controls, decoupling, and setup/hold margins.

Technical conclusion

The intended abstract decision is A AND B, optionally masked by Y. RP-style guarantees require a proven witness test and a deterministic precondition valid on all YES inputs. Independent trials accumulated by OR yield rejection error at most 2 to the power −k. Photon thinning requires the stated Bernoulli model. Neither nonclassical light nor CTC computation follows from the schematic alone.

References and provenance

1. User-supplied seven-page analysis, Optoelectronic RP Decision Circuit. Figure 1 is the original embedded image, reproduced without alteration. Its original labels remain conceptual.

2. S. Vadhan, The Power of Randomness, Definition 2.9 (RP). Harvard author manuscript: people.seas.harvard.edu/~salil/pseudorandomness/power.pdf.

3. D. Deutsch, “Quantum mechanics near closed timelike lines,” Physical Review D 44 (1991), 3197–3217.

4. S. Aaronson and J. Watrous, “Closed timelike curves make quantum and classical computing equivalent,” Proceedings of the Royal Society A 465 (2009), 631–647. Author manuscript: cs.uwaterloo.ca/~watrous/Papers/ClosedTimelikeCurves.pdf.

5. Photon-count identities follow from the moment definitions; the thinning relation follows from conditional binomial variance. Window probabilities follow from independent Bernoulli trials.